



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

IS2212-1 – BUSINESS INTELLIGENCE AND ITS APPLICATIONS
(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **QMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which type of data is defined by its tags and elements (metadata) but does not conform to a data model?
A) Structured data
B) Semi-structured data
C) Unstructured data
D) Binary data
2. Which of the following is a characteristic of structured data?
A) It does not conform to a data model.
B) Its definition, format, and meaning are explicitly known.
C) It is not easily usable by a program.
D) It is difficult to index and search.
3. Which type of system typically uses de-normalized tables with star or snowflake schema?
A) OLAP
B) OLTP
C) ERP
D) DSS
4. Who coined the term "BI" (Business Intelligence)?
A) Bill Gates
B) Howard Dresner
C) RN Prasad
D) Seema Acharya
5. Howard Dresner defined Business Intelligence as a set of concepts and methodologies to improve decision making through the use of:
A) Gut feelings and hunches.
B) Financial audits and reports.
C) Employee feedback.
D) Facts and fact-based systems.
6. Which of the following is NOT a synonym for Business Intelligence (BI)?
A) Business Analytics
B) Business Investment
C) Decision Support System
D) Competitive Intelligence
7. Which type of user uses different devices from desktop to smartphone to connect to enterprise IT applications?
A) Administrative users
B) Role based users
C) Knowledge workers
D) Multi-device access users
8. Which type of user configures the IT environment and manages users and access control?
A) Administrative users
B) Role based users
C) Knowledge workers
D) Multi-device access users
9. What is the primary purpose of the Program Management component in the BI framework?
A) To develop business applications.
B) To ensure seamless integration and smooth functioning of the entire program.
C) To handle data acquisition, cleaning, and integration.
D) To define business requirements.
10. What is one of the development challenges in Data Integration?
A) Unavailability of data.
B) Huge volumes of data.
C) Consistent and inconsistent metadata.
D) Manual integration risk.

11. What is a key purpose of a data staging area?
 - A) To perform final reporting.
 - B) To serve as an intermediate storage area for data from various sources.
 - C) To provide a final, unified view of the data to the user.
 - D) To exclusively store historical data.
12. Which phase of ETL involves converting and cleansing data before loading it into the target database?
 - A) Extract
 - B) Transform
 - C) Load
 - D) Publish
13. The Dimensional Modeling (DM) approach is primarily used for which type of system?
 - A) Online Transaction Processing (OLTP)
 - B) Online Analytical Processing (OLAP)
 - C) Enterprise Resource Planning (ERP)
 - D) Customer Relationship Management (CRM)
14. What is the operation of extracting data from a data source system called?
 - A) Data Loading
 - B) Data Transformation
 - C) Data Extraction
 - D) Data Cleansing
15. What is the primary purpose of the load phase in ETL?
 - A) To cleanse the data.
 - B) To transform the data.
 - C) To provide a history/audit trail of changes made to the data.
 - D) To load the data into the end target, usually a data warehouse.
16. Measures are a category of metrics that are:
 - A) Qualitative in nature.
 - B) Purely descriptive.
 - C) Quantitative measures, such as total sales.
 - D) Only used for operational purposes.
17. The number of responses to open job positions is an example of a KPI used by which department?
 - A) Sales
 - B) Marketing
 - C) Human Resources
 - D) Finance
18. What type of data is typically used in a Dashboard?
 - A) Detailed data
 - B) Summary data
 - C) Predictive data
 - D) Qualitative data
19. In the context of Cloud BI, what does IaaS stand for?
 - A) Internet as a Service
 - B) Integration as a Service
 - C) Information as a Service
 - D) Infrastructure as a Service
20. Which cloud service model is described as a "prepackaged BI solution"?
 - A) IaaS (Infrastructure as a Service)
 - B) PaaS (Platform as a Service)
 - C) SaaS (Software as a Service)
 - D) AaaS (Analytics as a Service)

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With an example, explain the key functions and core business processes in a modern organization.	10	L*2	1	1
	b) Differentiate between Structured, Unstructured, and Semi-Structured Data, providing an example for each.	10	L1	2	1
2.	a) Discuss the Role of OLAP Tools in the BI Architecture.	10	L3	1	1
	b) A company wishes to implement ERP and CRM systems—construct a case study showing how these enterprise applications transform information flow for different user roles.	10	L2	2	1
3.	a) Describe various information users in organizations and their typical requirements from IT applications, supported by a relevant case.	10	L2	1	1
	b) Compare and contrast OLTP and OLAP architectures, discussing their respective roles in BI systems.	10	L1	2	2

Unit – II

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| 4. a) | Describe the two principal components of a dimensional model: the Fact Table and the Dimension Table, and briefly discuss the Dimensional Modeling Life Cycle. | 10 | L1 | 3 | 2 |
| b) | A company, "FinCorp," is building a BI system. During the Transformation phase of their ETL process, they encounter Data Standardisation and Unit Conversion issues. Suggest specific techniques or rules that FinCorp can implement in their ETL transformation logic to resolve the Data Standardisation and Unit Conversion issues. | 10 | L2 | 4 | 2 |
| 5. a) | Discuss the BI Component Framework and prepare a scenario showing the involvement of various BI roles and responsibilities within a large organization. | 10 | L1 | 3 | 1 |
| b) | Define Data Integration and describe the core steps involved in the Extract, Transform, Load (ETL) process. | 10 | L2 | 4 | 2 |
| 6. a) | Explain the importance of Data Quality and Data Profiling in the context of data integration for Business Intelligence. | 10 | L2 | 4 | 1 |
| b) | What best practices should be followed in Business Intelligence and Data Warehousing (BIDW)? Support your answer with relevant situations. | 10 | L2 | 3 | 2 |

Unit – III

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| 7. a) | Describe how measures, metrics, and KPIs influence performance management in a manufacturing enterprise, using a case study. | 10 | L2 | 5 | 2 |
| b) | Describe the Enterprise Reporting Characteristics in the OLAP World. | 10 | L2 | 5 | 2 |
| 8. a) | Discuss the importance of standardized and well-presented enterprise reports at various organizational levels. | 10 | L2 | 5 | 1 |
| b) | The CEO of a manufacturing company requires a system to monitor the company's performance at a glance. They are considering implementing a Dashboard and a Balanced Scorecard. Identify and briefly discuss two trends from the BI Road Ahead that would impact the company's future reporting strategy. | 10 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
